THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why is the fermentation of dough a chemical change? 1

(a) dough freezes **(b)** dough melts

(c) gas bubbles and new substances form **(d)** dough dissolves

2. When sugar is heated strongly it turns black and gives off fumes. This is: 1

(a) a physical change **(b)** melting only

(c) a chemical change **(d)** a periodic change

3. The temperature of a mixture of water and quicklime rises because: 1

(a) water boils due to sunlight **(b)** ice melts

(c) a chemical reaction releases heat **(d)** evaporation

4. Which pair of changes shows both physical and chemical changes? 1

(a) freezing of water **(b)** melting of ice

(c) stretching of rubber **(d)** burning of a candle

5. Which of these is a sign that a chemical change has occurred when baking a cake? 1

(a) the batter rises and changes colour **(b)** the plate is clean

(c) the cake is round **(d)** the oven is hot

6. Why is the reaction between quicklime and water called a chemical change? 1

(a) lime floats **(b)** slaked lime, a new substance, forms

(c) water evaporates **(d)** lime melts

7. If the colour change on cutting a potato is stopped by putting it in water, it suggests: 1

(a) potato is a metal **(b)** water is a base

(c) reaction with air oxygen is slowed **(d)** potato dissolves

8. Which of these changes is not accompanied by the formation of a new substance? 1

(a) burning of a candle **(b)** digestion of food

(c) dissolving salt in water **(d)** fermentation of dough

9. **Assertion (A):** Galvanisation protects iron from rusting. 1

Reason (R): In galvanisation, iron is coated with a layer of copper.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** When baking soda is mixed with vinegar, the gas formed turns lime water milky. **Reason (R):** The gas formed is carbon dioxide. 1
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the shine of iron disappears on rusting. 2

12. Why does bread dough rise on keeping in a warm place? 2

13. Why is the tarnishing of silver a chemical change? 2

14. Why does a piece of burnt toast taste different from bread? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how a chemical change was used in making iron objects durable (galvanising) and in making steel. 3

16. Explain why heating of blue copper sulphate crystals produces water droplets in the cooler part. 3

17. Describe an experiment to show that heat is released in the reaction of quicklime with water. 3

OR

Why does a silver spoon turn black when kept in air for a long time? 3

18. Explain with a flow chart the changes in making butter from milk (curd, churning) and mark the type of each. 3

OR

Why is respiration a chemical change in our body? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In the kitchen, Mrs Khan notes four events: an ice cube melting in a glass, a slice of bread being toasted, sugar dissolving in tea and milk turning into curd overnight. 5
(i) Classify the melting of ice.
(ii) Classify the toasting of bread.
(iii) Classify the dissolving of sugar.
(iv) Classify the change of milk into curd and give two reasons for your answer.

20. Describe how the changes involved in fermentation of idli batter, baking of bread and curdling of milk are useful, and explain their chemical nature. 5

OR

Plan a short project to show that physical and chemical changes are involved in cooking five different dishes. 5

END OF QUESTION PAPER — Total: 40 Marks